

Roll No.

ODD END SEMESTER EXAMINATION DECEMBER - 2024

Name of the Course: B. Tech.

Semester: III

Name of the Paper: Fundamental of Cloud Computing and Big Data Paper Code: TCS 351

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b, & c in each main question.
- (iii) Total marks in each main question are Twenty.
- (iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	T-Systems, an Amazon Web Services (AWS) Partner, is using generative artificial intelligence (AI) tools including Amazon Bedrock to deliver cutting-edge solutions that combine AI and virtual reality. High performance, security, and cost-efficiency are vital. And that's why, says vice president of hyperscaler managed services Rachel Linnemann, the Deutsche Telekom subsidiary chooses to partner with AWS. Given a scenario where an organization is deciding between on-premises infrastructure and cloud computing, evaluate the advantages and disadvantages of cloud computing for their decision-making.	CO 1
(b)	Analyze the role of open standards in the evolution of cloud computing and discuss their importance in enabling interoperability and innovation in the digital age.	
(c)	Examine the five essential characteristics of cloud computing as defined by NIST.	
Q2	(20 marks)	
(a)	With data protection paramount, digital asset trading venue Rulematch works with Amazon Web Services (AWS) Partner Ergon to develop software securely in the cloud. Using AWS managed services helps Ergon find solutions fast to keep the clearing and settlement exchange running smoothly. In this video, Rulematch CEO David Riegel and Ergon software engineer Maximilian Falkenstein talk about the long-term partnership that's delivering best-in-class trading technology. Imagine you are designing a cloud application for a global e-commerce platform. How would the impact of network latency and web development choices influence the architecture and user interface (UI) of the platform?	CO 2
(b)	Compare and contrast the four cloud deployment models (Public, Private, Hybrid, and Community cloud). In what situations would each model be most suitable?	
(c)	Assess the impact of network infrastructure (such as bandwidth and latency) on the performance of cloud-based applications. How can organizations optimize their networks for better cloud performance?	
Q3	(20 marks)	
(a)	Capital One is now able to process checks up to 80% faster using AWS Step Functions Distributed Map, a visual workflow service for distributed applications. While most checks clear automatically, Capital One has a team of analysts who manually need to review checks with inconsistencies before the close of each business day. Design a hybrid cloud solution for a company	CO 3

	that handles sensitive customer data. Ensure that the design addresses security, scalability, and cost-effectiveness, combining both private and public cloud resources.	
(b)	Suppose you are managing a data center for an IaaS provider. How would you ensure scalability and resource optimization in the cloud, considering aspects like networking, storage, and VM provisioning?	
(c)	Compare and contrast the functionalities and use cases of Amazon Elastic Compute Cloud (EC2), Eucalyptus, and OpenStack. Which platform would you recommend for a company looking to implement an IaaS solution, and why?	
Q4	(20 marks)	
(a)	Mindtickle is a revenue enablement platform that helps ramp sales teams quickly, expand customer accounts, and win big. The Mindtickle platform keeps teams up to speed with market changes and buyers' needs, combining training, content management, coaching, call insights, and digital sales rooms in one. Mindtickle uses generative AI on Amazon Web Services (AWS) to power Mindtickle Copilot, to help sales teams prep for meetings faster, engage with the modern buyer, and close deals more efficiently. Given a scenario where a company wants to predict customer behavior, how would you apply machine learning techniques, such as regression analysis or sentiment analysis, to derive meaningful insights from their data?	CO 4
(b)	Analyze the advantages and disadvantages of using Big Data tools like association rule learning and classification tree analysis. In what situations would each tool be most effective?	
(c)	Design a Big Data analytics solution for a retail company looking to optimize inventory and predict future demand. Incorporate tools like data fusion, A/B testing, and machine learning techniques. What would be the key steps in developing this solution?	
Q5	(20 marks)	
(a)	Examine the role of AWS services such as Amazon CloudFront and Amazon Glacier in content delivery and data storage. How do these services enhance scalability and performance for cloud applications?	CO5 &
(b)	If you were responsible for securing a cloud-based application, how would you use AWS security services to ensure compliance with industry standards and protect against data breaches?	CO6
(c)	Contentful uses Amazon Web Services (AWS) generative artificial intelligence (AI) capabilities in Amazon Bedrock to deliver the flexibility its diverse client base demands. The company's in-platform AI features change the way personalized content is created, making it easy for customers, including numerous Fortune 500 organizations, to hit the right tone for their target audiences. Conduct a SWOT analysis of AWS cloud services, focusing on factors such as savings, scalability, and reliability. How do AWS services provide a value proposition for businesses seeking to migrate to the cloud?	